

Supplementary Information

A Related Work

LLMs are getting huge interest as a tool for applications on various educational scenarios (1). However, recent studies show that current LLMs fail to output grade-appropriate answers (2) regardless of how informative the prompt is (3). Current models consistently default to a high school (4) or college+ level (5), even when explicitly prompted to target a lower grade level. The root of this challenge lies in the training data itself, as foundational datasets such as Alpaca and Dolly exhibit a strong bias towards college+ level (6).

Beyond prompts, prior research has explored fine-tuning to align with different grade-levels. However, these efforts are confined to text simplification tasks that require a source text (e.g., summarization and paraphrasing) (7; 8) and have shown poor performance for lower grades (9). Furthermore, the works have been limited to incorporating only a few educational metrics, given the difficulty of optimizing for multiple criteria simultaneously. In contrast, our work targets open-ended question answering, simulating a more realistic classroom setting of direct student-LLM interaction. By incorporating seven educational metrics, we achieve strong performance across all grade-levels, proving effective even for the lower-elementary level.

B Metrics

B.1 Educational Metrics

We compute text readability using multiple readability metrics. As each metric has different methods for mapping scores to grade levels, we standardize the results by classifying them into six predefined grade levels: 1 (lower elementary), 2 (middle elementary), 3 (high elementary), 4 (middle), 5 (high), and 6 (adult).

B.1.1 Flesch Reading Ease

The Flesch reading-ease score (FRES) (10), used by the U.S. Department of Defense, provides a numerical measure of text readability:

$$\text{FRES} = 206.835 - 1.015 \times \left(\frac{\text{total words}}{\text{total sentences}} \right) - 84.6 \times \left(\frac{\text{total syllables}}{\text{total words}} \right).$$

Higher FRES values correspond to text that is easier to read.

B.1.2 Flesch–Kincaid grade level

The Flesch–Kincaid Grade Level (FKGL) (11) formula is widely employed in education to estimate the U.S. grade level at which a text can be understood. It is calculated as:

$$\text{FKGL} = 0.39 \times \left(\frac{\text{total words}}{\text{total sentences}} \right) + 11.8 \times \left(\frac{\text{total syllables}}{\text{total words}} \right) - 15.59.$$

Higher FKGL scores correspond to texts requiring more advanced reading skills.

B.1.3 Coleman–Liau Index

The Coleman–Liau Index (CLI) (12) quantifies text readability by analyzing the average number of letters and sentences per 100 words. It is given by:

$$\text{CLI} = 0.0588 \times L - 0.296 \times S - 15.8,$$

where L is the average number of letters per 100 words and S is the average number of sentences per 100 words. Higher CLI scores correspond to texts requiring more advanced reading skills.

B.1.4 Linsear Write

Linsear Write (LW) (13) is a readability test developed for the U.S. Air Force to evaluate how easily technical manuals can be read. It is computed over a 100-word passage as follows:

1. Let E be the number of “easy” words (words with at most two syllables), and H be the number of “hard” words (three or more syllables).
2. Let S be the number of sentences in the 100-word sample.
3. Compute the intermediate score i :

$$i = \frac{E + 3H}{S}.$$

4. Determine the final Linsear Write score (LW):

$$\text{LW} = \begin{cases} \frac{i}{2}, & \text{if } i > 20, \\ \frac{i-2}{2}, & \text{otherwise.} \end{cases}$$

Higher LW scores correspond to texts requiring more advanced reading skills.

B.1.5 Gunning–Fog Index

The Gunning–Fog Index (Fog) (14) estimates the years of formal education required to comprehend a passage on the first reading. This index is defined as:

$$\text{Fog} = 0.4 \left(\frac{W}{S} + 100 \frac{C}{W} \right).$$

where W is the total number of words, S is the total number of sentences, and C is the number of “complex” words with three or more syllables. Higher Fog values indicate text requiring more advanced reading skills.

B.1.6 Dale–Chall Readability Formula

The Dale–Chall readability formula (DC) (15) assesses text complexity based on the proportion of unfamiliar words and sentence length, using a predefined list of 3000 words familiar to 80% of fifth-grade students. It is defined as:

$$\text{DC}_{\text{raw}} = 0.1579p + 0.0496 \frac{W}{S}.$$

The final (adjusted) score is:

$$\text{DC} = \begin{cases} \text{DC}_{\text{raw}} + 3.6365, & \text{if } p > 5, \\ \text{DC}_{\text{raw}}, & \text{otherwise.} \end{cases}$$

where D is the number of difficult words (words not in the predefined list), W is the total number of words, S is the total number of sentences, and p is the percentage of difficult words ($\frac{D}{W} \times 100$). Higher DC values indicate text requiring more advanced reading skills.

B.1.7 Spache Readability Formula

The Spache readability formula (Sp) (16) tests text readability for younger children. It uses a predefined list of approximately 1000 words and is defined as:

$$\text{Sp} = 0.141 S + 0.086 U + 0.839.$$

where S is the average sentence length (words per sentence) and U is the percentage of unique unfamiliar words (words not in the predefined list). Higher Sp values indicate text requiring more advanced reading skills.

B.1.8 Automated Readability Index

The Automated Readability Index (ARI) (17) estimates the U.S. grade level required to comprehend a text. It is defined as:

$$\text{ARI} = 4.71 \frac{C}{W} + 0.5 \frac{W}{S} - 21.43.$$

where C is the total number of characters (letters and digits), W is the total number of words, and S is the total number of sentences. Higher ARI scores indicate text requiring more advanced reading skills.

B.2 Perplexity

Perplexity (ppl) predicts the probability of a token w_i given its preceding context $w_1, \dots, w_{i-1}$. The perplexity is defined as:

$$\text{ppl} = \exp \left(-\frac{1}{N} \sum_{i=1}^N \log p(w_i \mid w_1, \dots, w_{i-1}) \right),$$

where N is the number of tokens in the text dataset with log as the natural logarithm. Higher perplexity implies a broader probability spread across potential outcomes, indicating more diverse outputs. We use GPT2-XL for computing the probability.

B.3 Diversity Gain

Diversity gain (18) quantifies the extent to which a generated dataset brings diversity to the original (base) dataset. The base dataset is defined as

$$D_{\text{base}} = \{x_i = (q_i, r_i, a_i)\}_{i=1}^N$$

with N samples. The new generated dataset is defined as

$$D_{\text{new}} = \{x_i = (q_i, r_i, a_i)\}_{i=1}^M$$

with M samples. The diversity gain of D_{new} relative to D_{base} is

$$d_{\text{gain}} = \frac{1}{M} \sum_{x_i \in D_{\text{new}}} \min_{x_j \in D_{\text{base}}} \|f(x_i) - f(x_j)\|,$$

where f is the feature extractor. We use D_{base} as the collection of text outputs from the base model (GPT4o-mini) before finetuning and D_{new} as the collection of text outputs from the grade-level finetuned models for all four datasets. We use `text-embedding-3-large` from OpenAI to extract the features.

C Extended Results

C.1 Fields and Subjects for Training Text Data

We present the fields and subjects of questions used for finetuning based on the K-12 curriculum.

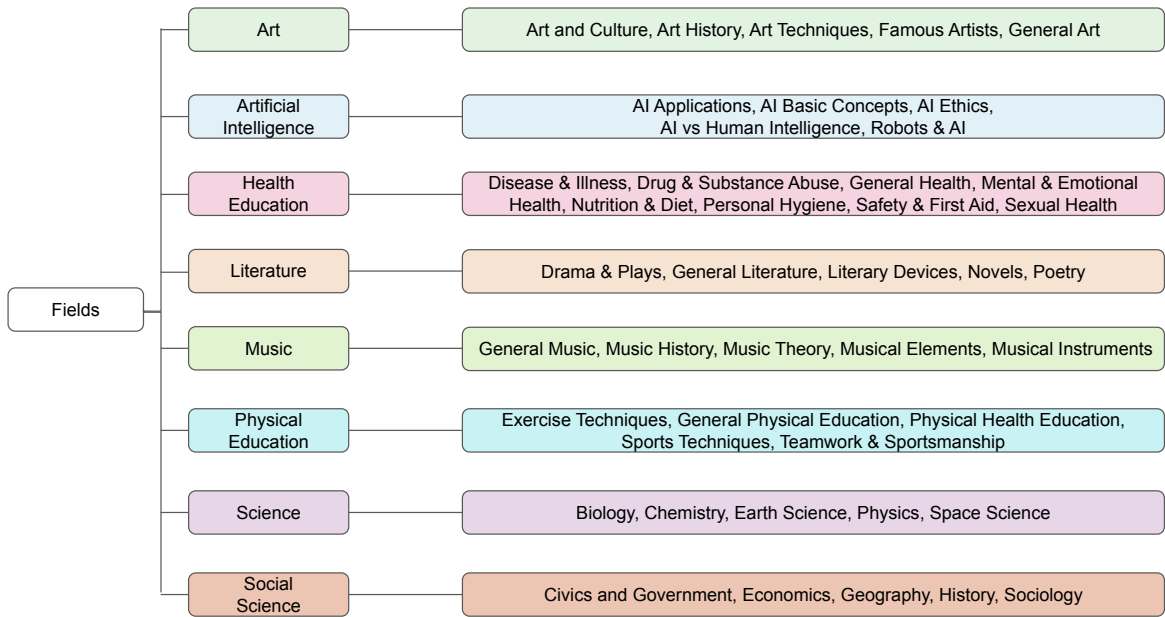


Figure 1: Fields and subjects of questions used for finetuning.

C.2 Number of Q&A for each Grade-level

Table 1 provides a detailed distribution of question & answer pairs across the six grade-levels. We collect a total of 1,287,149 pairs. The amount of data for lower elementary is comparatively smaller than those for other grade-levels, due to the relatively strict thresholds set by the educational metrics. However, the absolute number of data remains substantial enough to facilitate effective finetuning.

L-Elem	M-Elem	H-Elem	Middle	High	Adult	Total
71,524	324,910	302,485	275,131	130,779	182,590	1,287,149

Table 1: Number of question & answer pairs used for finetuning different models.

C.3 Extended Output Distribution

We compare the output grade-level distributions from the prompt-based approach and ours in Figure 2. Overall, our approach shows a diagonal pattern, indicating stronger alignment with the intended grade levels than the prompt-based approach.

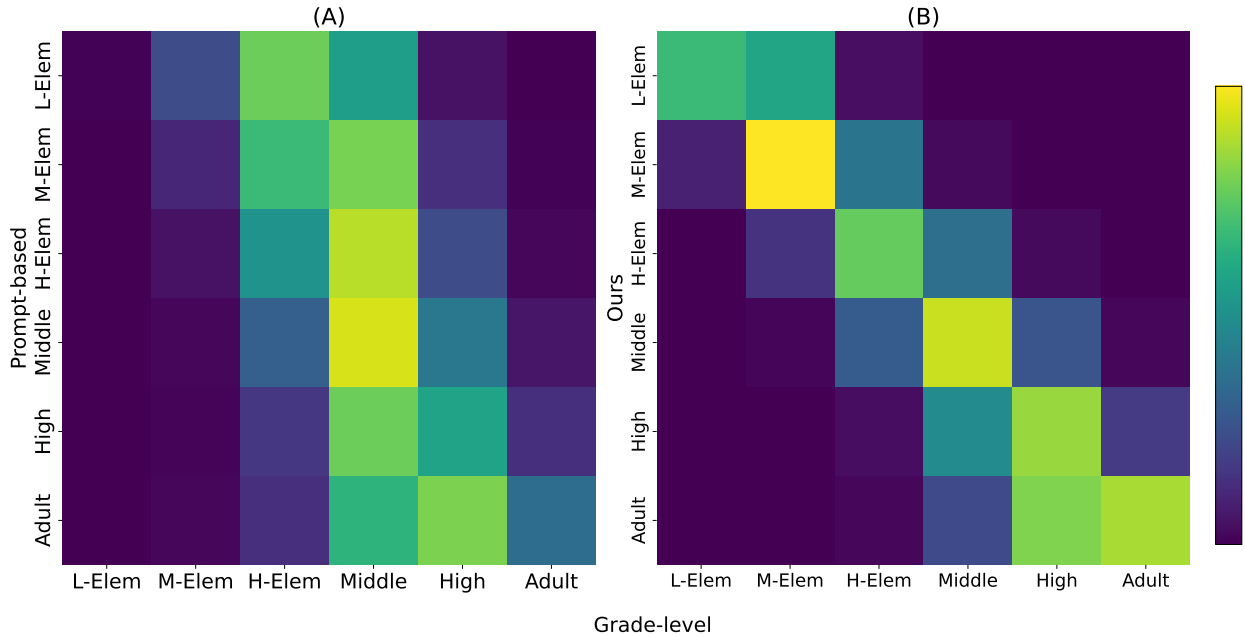


Figure 2: Comparison of output grade-level distributions from the prompt-based approach (A) and our method (B). The rows represent the targeted grade-level (prompt or model), and the columns show the classified grade-levels. Warmer colors (e.g., yellow) represent higher frequencies. For instance, the top row of (B) reveals the output classification results of the lower elementary model’s responses.

C.4 Text Distribution in Data Generation for Different Prompts

We show the output grade-level distribution across different prompts, which are used for data generation. No single prompt consistently aligns with the intended grade level, highlighting the need for our proposed finetuning method.

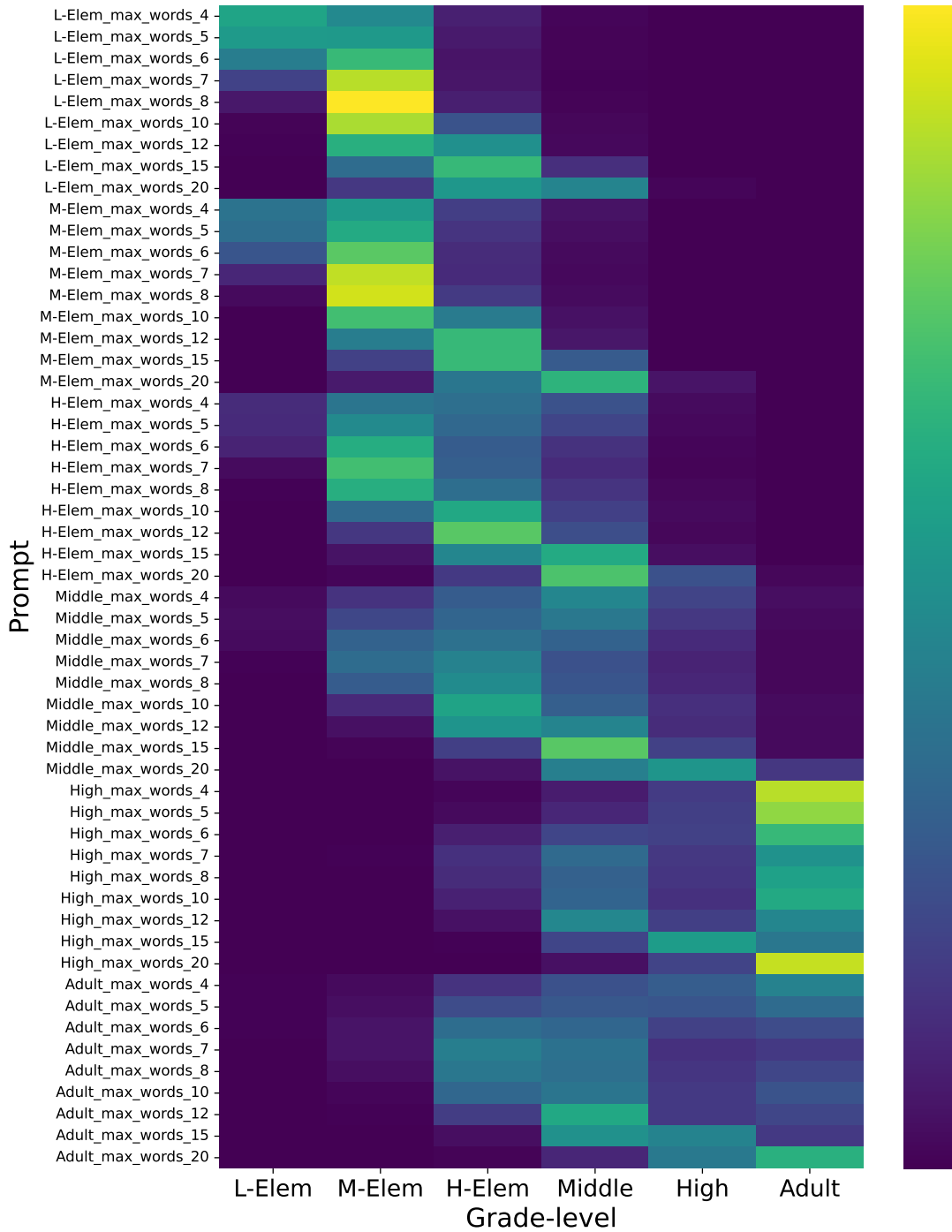


Figure 3: Grade distribution for different prompts in data generation. Each row represents a specific prompt, while the columns indicate the classified grade levels.

C.5 Compatibility Results for Base LLMs

In Table 2, we compare the grade-level distribution of outputs among different base LLMs and their ARI scores. The results show that the current state-of-the-art LLMs tend to output with higher-grade readability, highlighting the challenge of generating content suitable for lower grade students.

Model	Dataset	L-Elem (1)	M-Elem (2)	H-Elem (3)	Middle (4)	High (5)	Adult (6)	ARI
GPT4o-mini	$\mathcal{D}_{GPT}$	0.00	0.00	7.57	44.59	34.73	13.11	5.25
	$\mathcal{D}_{SQ2}$	0.67	3.99	8.02	31.53	43.81	14.74	4.21
	$\mathcal{D}_{ELI}$	0.00	0.08	1.59	37.93	49.26	11.14	5.24
	$\mathcal{D}_{NQ}$	0.03	1.14	11.00	35.69	36.78	15.37	4.82
GPT4o	$\mathcal{D}_{GPT}$	0.00	0.27	7.16	41.35	35.27	15.95	X
	$\mathcal{D}_{SQ2}$	1.15	3.09	10.96	32.91	45.51	13.63	X
	$\mathcal{D}_{ELI}$	0.00	0.08	1.67	38.09	47.49	12.67	X
	$\mathcal{D}_{NQ}$	0.07	1.33	11.55	36.86	35.84	14.34	X
LLaMA 3.3:70B	$\mathcal{D}_{GPT}$	0.00	0.68	11.89	46.35	32.16	8.92	5.06
	$\mathcal{D}_{SQ2}$	0.54	3.10	12.58	27.58	47.29	13.26	4.87
	$\mathcal{D}_{ELI}$	0.00	0.06	1.99	30.21	50.26	17.47	5.33
	$\mathcal{D}_{NQ}$	0.12	0.96	8.35	35.37	40.83	14.38	4.98
LLaMA 3.1:70B	$\mathcal{D}_{GPT}$	0.00	0.14	13.38	51.35	28.51	6.62	5.04
	$\mathcal{D}_{SQ2}$	1.16	2.72	10.16	26.73	44.47	16.41	4.92
	$\mathcal{D}_{ELI}$	0.00	0.08	2.44	33.64	48.51	15.32	5.35
	$\mathcal{D}_{NQ}$	0.15	0.86	7.95	34.60	41.20	15.24	4.98
LLaMA 3.1:8B	$\mathcal{D}_{GPT}$	0.00	0.41	10.95	48.92	32.16	7.57	5.09
	$\mathcal{D}_{SQ2}$	0.92	2.65	11.45	31.67	44.76	12.82	4.80
	$\mathcal{D}_{ELI}$	0.00	0.06	2.27	33.11	48.89	15.68	5.33
	$\mathcal{D}_{NQ}$	0.12	0.65	7.47	35.38	42.05	14.34	4.98
Qwen 2.5:72B	$\mathcal{D}_{GPT}$	0.00	0.00	6.22	39.32	42.03	12.43	5.24
	$\mathcal{D}_{SQ2}$	0.25	3.30	10.44	31.98	49.78	9.60	4.85
	$\mathcal{D}_{ELI}$	0.00	0.03	1.30	33.83	53.59	11.25	5.28
	$\mathcal{D}_{NQ}$	0.01	0.27	5.76	33.67	44.16	16.13	5.07
Gemma 2:27B	$\mathcal{D}_{GPT}$	0.00	0.41	8.24	29.73	37.43	24.19	5.29
	$\mathcal{D}_{SQ2}$	3.73	8.61	22.40	35.43	34.88	9.53	4.42
	$\mathcal{D}_{ELI}$	0.01	0.19	2.67	28.87	45.03	23.23	5.37
	$\mathcal{D}_{NQ}$	0.36	9.07	21.81	31.78	24.43	12.56	4.25
Phi 4:14B	$\mathcal{D}_{GPT}$	0.00	0.00	4.87	36.89	42.16	13.24	5.44
	$\mathcal{D}_{SQ2}$	0.00	1.63	7.12	26.65	40.76	16.66	5.16
	$\mathcal{D}_{ELI}$	0.00	0.00	0.83	23.39	58.65	17.13	5.58
	$\mathcal{D}_{NQ}$	0.03	0.70	8.56	30.44	39.13	21.15	5.04
Mixtral 8x7B	$\mathcal{D}_{GPT}$	0.00	1.36	15.20	37.04	29.44	16.96	5.07
	$\mathcal{D}_{SQ2}$	2.33	2.95	14.40	36.30	44.15	9.79	4.72
	$\mathcal{D}_{ELI}$	0.01	0.16	4.60	31.31	44.97	18.95	5.24
	$\mathcal{D}_{NQ}$	0.14	0.93	10.39	36.81	39.22	12.52	4.88

Table 2: Target % and ARI values for various base LLMs across four datasets. Each column (lower elementary to adult) shows the proportion of outputs classified at that grade level, while the final column presents the ARI value. Bolded entries indicate the highest proportion for each model and dataset pair.

C.6 Compatibility Results for Prior Approaches and Ours

We provide a comprehensive breakdown of grade-level output distribution for several models across four datasets, comparing prompt-based approach and ours. Table 3 includes both the percentage of outputs correctly aligned with each target grade level (Target %) and the ARI values. Notably, our finetuned models often achieve stronger alignment at the targeted grade-levels.

	Model	Dataset	L-Elem (1)		M-Elem (2)		H-Elem (3)		Middle (4)		High (5)		Adult (6)	
			Target %	ARI	Target %	ARI	Target %	ARI	Target %	ARI	Target %	ARI	Target %	ARI
Prompt Based	GPT4o mini (Baseline)	$\mathcal{D}_{GPT}$	0.13	3.35	11.22	3.77	36.22	3.99	50.68	4.36	27.30	4.87	22.16	5.40
		$\mathcal{D}_{SQ2}$	8.04	2.67	16.80	3.40	37.28	3.84	54.30	4.48	47.32	5.10	21.95	5.13
		$\mathcal{D}_{ELI}$	0.05	3.51	5.36	3.77	37.99	3.96	64.18	4.32	32.58	4.82	14.96	5.35
		$\mathcal{D}_{NQ}$	0.25	3.29	5.38	3.75	25.42	4.07	49.94	4.40	35.55	4.66	18.00	4.90
	GPT4o	$\mathcal{D}_{GPT}$	0.27	3.39	14.46	3.61	31.62	3.89	52.70	4.26	35.14	4.96	21.62	5.31
		$\mathcal{D}_{SQ2}$	4.70	2.98	14.17	3.60	32.33	3.95	52.39	4.51	48.74	5.14	16.95	5.04
		$\mathcal{D}_{ELI}$	0.13	3.49	8.71	3.63	41.29	3.83	62.38	4.21	26.47	4.60	13.06	5.22
		$\mathcal{D}_{NQ}$	1.20	2.97	14.31	3.36	34.09	3.78	56.67	4.22	35.34	4.67	15.53	4.92
	LLaMA 3.3:70B	$\mathcal{D}_{GPT}$	0.14	3.22	10.81	3.65	31.89	4.04	48.51	4.44	29.19	4.81	12.57	5.19
		$\mathcal{D}_{SQ2}$	6.10	2.48	17.69	3.38	34.50	3.89	47.81	4.68	49.86	5.16	14.66	5.06
		$\mathcal{D}_{ELI}$	0.02	3.54	2.56	3.92	18.30	4.29	53.89	4.73	47.74	5.11	18.87	5.35
		$\mathcal{D}_{NQ}$	0.59	3.31	6.04	3.86	17.73	4.34	44.25	4.72	41.80	4.93	14.53	5.08
	LLaMA 3.1:70B	$\mathcal{D}_{GPT}$	0.27	3.20	10.68	3.69	34.60	4.03	53.51	4.39	24.73	4.75	9.19	5.10
		$\mathcal{D}_{SQ2}$	5.37	2.49	18.98	3.35	31.09	3.96	49.43	4.65	46.57	4.96	11.74	4.97
		$\mathcal{D}_{ELI}$	0.03	3.52	2.76	3.87	19.89	4.29	56.10	4.70	43.50	5.06	13.99	5.30
		$\mathcal{D}_{NQ}$	0.66	3.28	5.40	3.90	16.32	4.40	45.08	4.67	39.04	4.84	15.45	5.07
	LLaMA 3.1:8B	$\mathcal{D}_{GPT}$	0.14	3.26	13.67	3.58	40.96	3.92	46.22	4.35	27.70	4.83	10.95	5.18
		$\mathcal{D}_{SQ2}$	6.80	2.50	23.76	3.17	38.69	3.78	49.52	4.48	44.19	4.91	10.21	4.85
		$\mathcal{D}_{ELI}$	0.02	3.49	4.10	3.77	24.85	4.16	57.44	4.64	43.17	5.04	12.85	5.27
		$\mathcal{D}_{NQ}$	0.49	3.32	6.10	3.82	17.98	4.31	45.94	4.61	39.77	4.86	14.12	5.06
	Qwen 2.5:72B	$\mathcal{D}_{GPT}$	0.00	3.59	9.87	3.70	42.97	3.80	63.38	4.43	37.97	5.07	13.24	5.27
		$\mathcal{D}_{SQ2}$	4.16	2.92	19.10	3.33	45.65	3.62	59.90	4.42	51.51	5.03	9.81	4.86
		$\mathcal{D}_{ELI}$	0.03	3.57	5.19	3.68	40.64	3.87	68.9	4.44	49.29	5.10	8.88	5.22
		$\mathcal{D}_{NQ}$	0.10	3.52	5.07	3.80	20.16	4.28	47.16	4.67	43.34	4.97	15.57	5.03
	Gemma 2:27B	$\mathcal{D}_{GPT}$	1.14	2.97	17.16	3.36	41.76	3.64	54.73	3.94	25.14	4.55	29.46	5.38
		$\mathcal{D}_{SQ2}$	15.53	1.92	37.91	2.91	47.49	3.28	46.32	3.74	28.12	4.47	12.65	4.65
		$\mathcal{D}_{ELI}$	0.13	3.19	11.99	3.53	43.18	3.76	59.49	4.11	27.61	4.71	25.50	5.42
		$\mathcal{D}_{NQ}$	7.58	2.34	31.97	2.77	40.49	3.08	40.26	3.58	21.74	4.02	12.70	4.22
	Phi 4:14B	$\mathcal{D}_{GPT}$	0.00	3.72	9.87	3.86	39.46	4.02	55.27	4.35	31.49	5.07	12.57	5.43
		$\mathcal{D}_{SQ2}$	1.71	3.05	16.12	3.49	39.95	3.81	58.25	4.52	48.29	5.20	12.65	5.07
		$\mathcal{D}_{ELI}$	0.00	3.84	3.48	3.93	35.27	4.05	67.16	4.47	46.45	5.18	12.56	5.51
		$\mathcal{D}_{NQ}$	0.10	3.72	3.47	4.05	17.51	4.32	45.15	4.77	47.91	5.20	23.06	5.17
	Mixtral :8x7B	$\mathcal{D}_{GPT}$	0.68	3.83	7.57	3.97	18.02	4.62	50.81	4.54	34.39	4.92	21.94	5.22
		$\mathcal{D}_{SQ2}$	1.96	3.16	10.60	3.87	15.96	4.55	45.54	4.61	45.73	5.03	11.17	4.75
		$\mathcal{D}_{ELI}$	0.02	4.28	2.00	4.41	4.81	5.10	44.56	4.93	48.85	5.25	20.65	5.35
		$\mathcal{D}_{NQ}$	0.09	3.96	2.16	4.46	7.01	4.94	41.49	4.81	44.66	5.04	13.32	4.95
Ours	Finetuned GPT4o mini	$\mathcal{D}_{GPT}$	67.30	1.32	82.97	2.14	73.11	3.42	70.54	4.56	63.24	5.33	73.78	5.40
		$\mathcal{D}_{SQ2}$	65.99	1.21	67.08	2.20	52.74	3.56	58.83	4.45	65.40	5.26	43.16	5.67
		$\mathcal{D}_{ELI}$	57.41	1.36	72.82	2.24	60.00	3.55	65.18	4.38	64.68	5.34	66.16	5.88
		$\mathcal{D}_{NQ}$	46.65	1.37	61.90	2.07	58.21	3.27	58.71	4.23	49.07	4.99	33.54	5.49

Table 3: Extended performance comparison across multiple models, grade levels, and datasets. Each cell displays the Target% and the corresponding ARI value. Bold values indicate the highest performance for a particular dataset-grade combination.

C.7 Survey Results with GPT

We present the results on the survey results from GPT4o. We feed GPT4o with the same set of survey questions posed to human participants. For type 1 questions, we observe Kendall’s τ coefficient as 0.72 and an L1 distance of [0.419, 0.585, 0.635, 0.831, 0.724, 0.601], along with a strong alignment between model outputs and GPT4o’s perceptions (see Figure 4). For type 2 questions, answer comprehensibility and accuracy consistently remains high regardless of the question difficulty (see Figure 5). These high correlation between GPT4o’s output ratings and the ratings provided by human participants robustly validates our framework’s effectiveness.

C.7.1 Type 1 Question

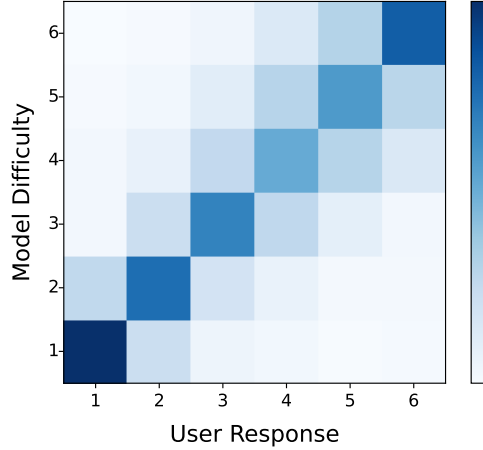


Figure 4: Survey results for type 1 questions by GPT4o.

C.7.2 Type 2 Question

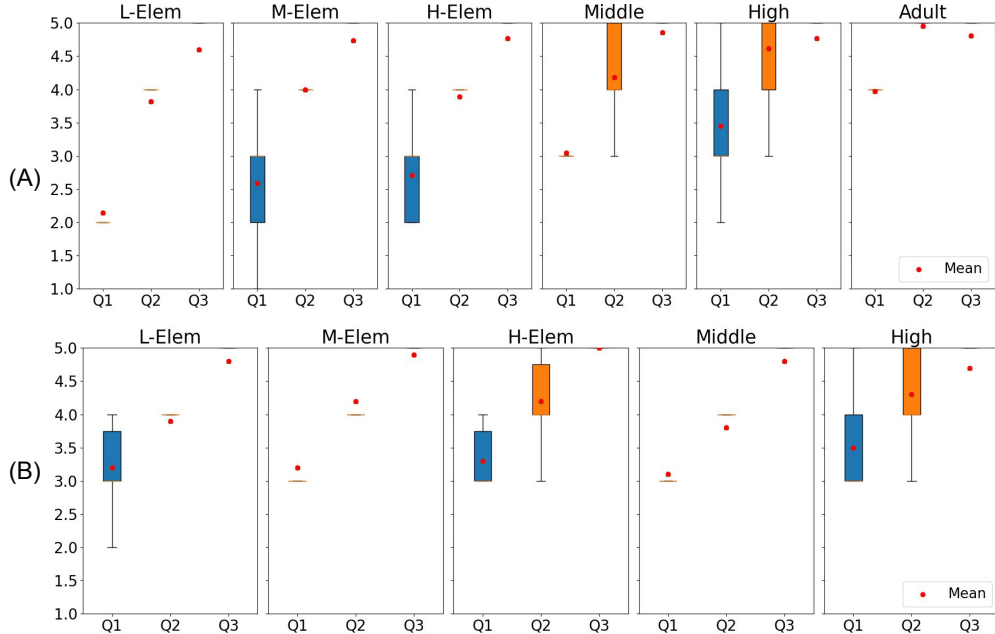


Figure 5: Survey results for type 2 questions by GPT4o.

D Exploring Worldview of Models–Output Distribution

We analyze the different worldviews of models through output analysis across the experimental datasets. Since educational metrics primarily capture sentence and word-level complexity, we present three analyses of these textual aspects.

D.1 Sentence Length Distribution

As shown in Figure 6, sentence length increases for higher-grade models. These models produce more detailed explanations that require advanced reading skills. Lower grade models generate shorter statements that match younger learners’ language needs (19). Each model’s worldview shifts with its target grade level, from concise simplicity to detailed explanation.

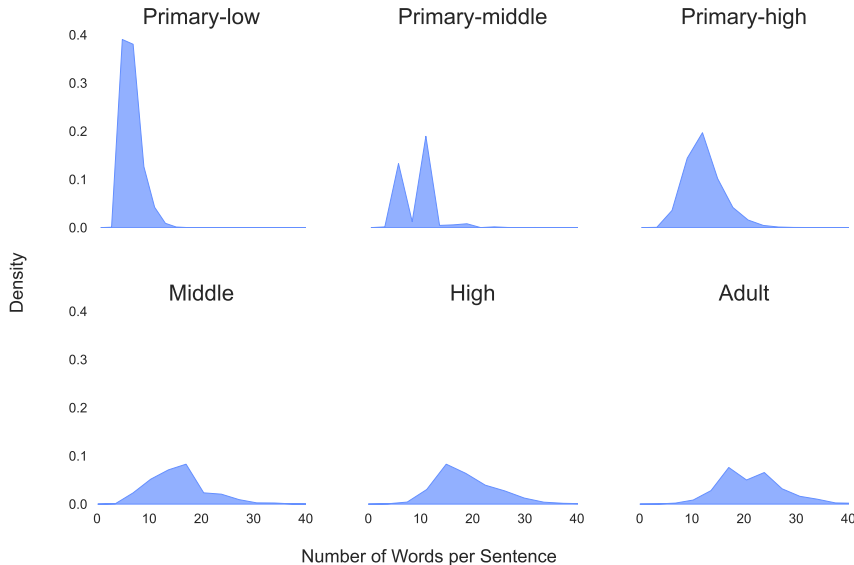


Figure 6: Sentence length distribution across different models.

D.2 Word Difficulty

A key component of model worldview analysis involves examining output vocabulary. Humans use different words as they develop, even when describing identical situations. We visualize outputs through word clouds and Zipfian distributions (20), applying BM25 (21) to highlight distinctive terms while minimizing common words like “people” and “like.” We exclude words appearing fewer than 30 times to eliminate outliers. Figure 7 reveals vocabulary shifts across different models, with higher-grade models using more complex terms that reflect advanced linguistic development.

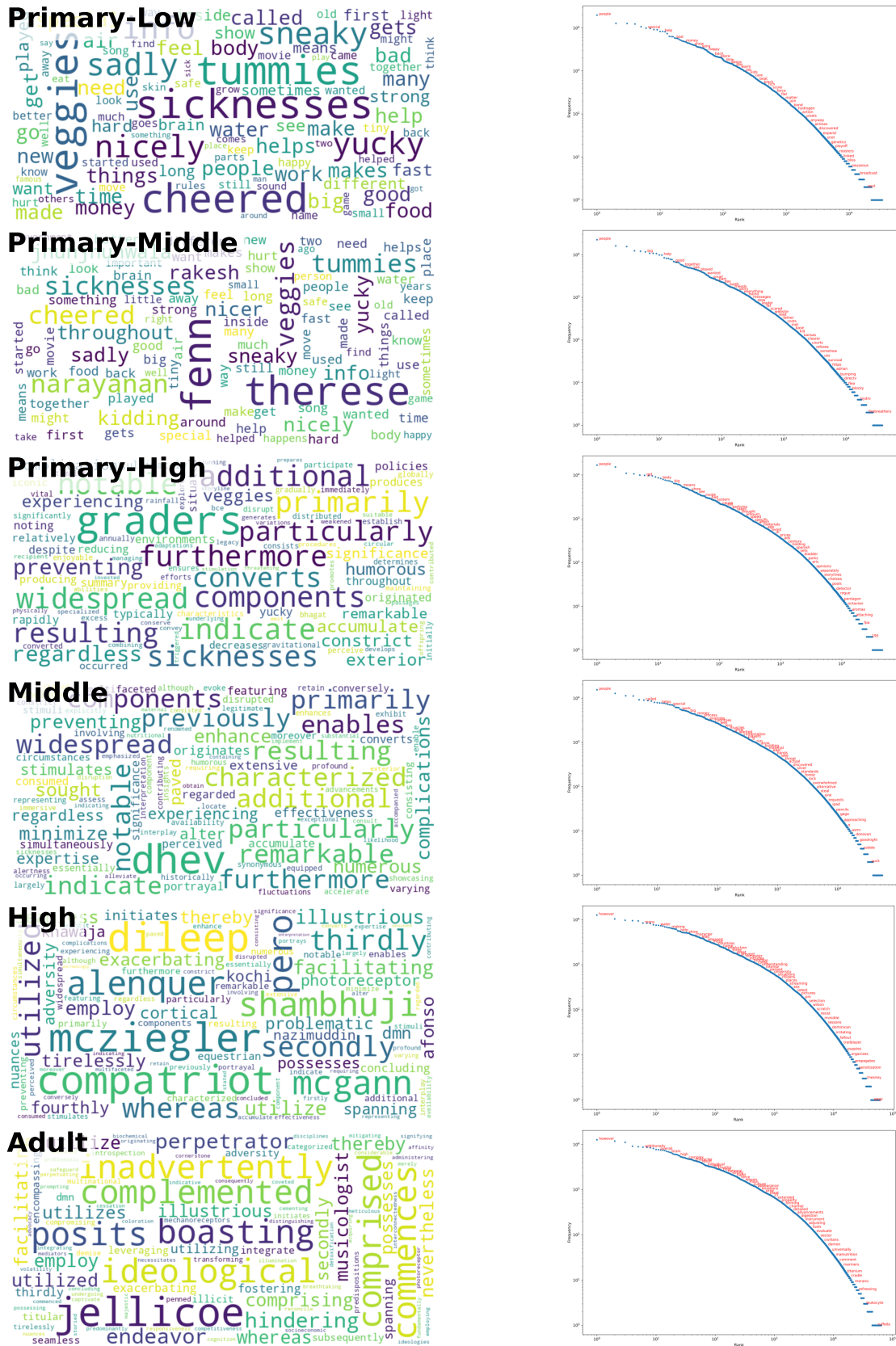


Figure 7: Word Cloud and Zipfian Distribution

E Survey Details

We provide further details on how we distribute and design the surveys.

E.1 Survey Distribution Details

For Survey 1, we sampled questions from $\mathcal{D}_{NQ}$, consisting of real user questions submitted to Google Search. We assessed whether finetuned model answers to natural questions were both comprehensible for target grades and factually accurate. We provided 5 Type 1 questions and 18 Type 2 questions to each participant with compensation of 7000 Korean Won (approximately 5 USD) for an estimated 30 minutes of work. We recruited 108 English-speaking participants from the Republic of Korea, the USA, and England. To ensure question diversity, we created 18 distinct forms, each completed by six participants.

For Survey 2, we sampled questions from $\mathcal{D}_{SQ2}$ to measure the comprehensibility and accuracy of finetuned models for grade-appropriate questions. $\mathcal{D}_{SQ2}$ contains questions classified for grades 1-12 based on difficulty. We provided 10 Type 2 questions per grade level to 100 participants from Amazon Mechanical Turk. Participants received 1.5 USD for an estimated 8 minutes of work.

E.2 Type 1 Question

- Q: Assign each answer to a unique grade level based on how comprehensible/understandable it would be to students of that grade.

E.3 Type 2 Questions

- Q1: Is the question answerable by X-grade students (regardless of the given answer)?
 - 5: Any X-grade student can answer this question.
 - 4: Most X-grade students can answer this question.
 - 3: It is likely that X-grade students can answer this question.
 - 2: Most X-grade students cannot answer this question.
 - 1: It is impossible for X-grade students to answer this question.
- Q2: Is the answer comprehensible by X-grade students (regardless of the given question)?
 - 5: Any X-grade student can comprehend this answer.
 - 4: Most X-grade students can comprehend this answer.
 - 3: It is likely that X-grade students can comprehend this answer.
 - 2: Most X-grade students cannot comprehend this answer.
 - 1: It is impossible for X-grade students to comprehend this answer.
- Q3: Is the answer consistently answering the question?
 - 5: The answer answers the corresponding question.
 - 4: The answer mostly follows the corresponding question.
 - 3: The answer roughly follows the corresponding question.
 - 2: The answer hardly follows the corresponding question.
 - 1: The answer is completely irrelevant to the corresponding question.

References

- [1] Wang, S. *et al.* Large language models for education: A survey and outlook (2024). URL <https://arxiv.org/abs/2403.18105>. 2403.18105.
- [2] Gobara, S., Kamigaito, H. & Watanabe, T. Do llms implicitly determine the suitable text difficulty for users? (2024). URL <https://arxiv.org/abs/2402.14453>. 2402.14453.
- [3] Imperial, J. M. & Tayyar Madabushi, H. Flesch or fumble? evaluating readability standard alignment of instruction-tuned language models. In *Proceedings of the Third Workshop on Natural Language Generation, Evaluation, and Metrics (GEM)*, 205–223 (Association for Computational Linguistics, Singapore, 2023). URL <https://aclanthology.org/2023.gem-1.18/>.
- [4] Hsu, Y.-S., Feldhus, N. & Hakimov, S. Free-text rationale generation under readability level control. *ArXiv abs/2407.01384* (2024). URL <https://api.semanticscholar.org/CorpusID:270870139>.
- [5] Rooein, D., Curry, A. C. & Hovy, D. Know your audience: Do llms adapt to different age and education levels? (2023). URL <https://arxiv.org/abs/2312.02065>. 2312.02065.
- [6] Roegiest, A. & Pinkosova, Z. Generative information systems are great if you can read. In *Proceedings of the 2024 Conference on Human Information Interaction and Retrieval, CHIIR '24*, 165–177 (Association for Computing Machinery, New York, NY, USA, 2024). URL <https://doi.org/10.1145/3627508.3638345>.
- [7] Malik, A., Mayhew, S., Piech, C. & Bicknell, K. From tarzan to Tolkien: Controlling the language proficiency level of LLMs for content generation. In *Findings of the Association for Computational Linguistics: ACL 2024*, 15670–15693 (Association for Computational Linguistics, Bangkok, Thailand, 2024). URL <https://aclanthology.org/2024.findings-acl.926/>.
- [8] Ribeiro, L. F. R., Bansal, M. & Dreyer, M. Generating summaries with controllable readability levels. In *The 2023 Conference on Empirical Methods in Natural Language Processing* (2023). URL <https://openreview.net/forum?id=IFNbElsnCi>.
- [9] Tran, H., Yao, Z., Li, L. & Yu, H. Readctrl: Personalizing text generation with readability-controlled instruction learning (2024). URL <https://arxiv.org/abs/2406.09205>. 2406.09205.
- [10] Flesch, R. A new readability yardstick. *Journal of applied psychology* **32**, 221 (1948).
- [11] Kincaid, J. Derivation of new readability formulas (automated readability index, fog count and flesch reading ease formula) for navy enlisted personnel. *Chief of Naval Technical Training* (1975).
- [12] Coleman, M. & Liau, T. L. A computer readability formula designed for machine scoring. *Journal of Applied Psychology* **60**, 283 (1975).
- [13] O’hayre, J. *Gobbledygook has gotta go* (US Department of the Interior, Bureau of Land Management, 1966).
- [14] Gunning, R. The technique of clear writing (1952).
- [15] Chall, J. S. & Dale, E. Readability revisited: The new dale-chall readability formula. (*No Title*) (1995).
- [16] Spache, G. A new readability formula for primary-grade reading materials. *The Elementary School Journal* **53**, 410–413 (1953).
- [17] Smith, E. A. & Senter, R. *Automated readability index*, vol. 66 (Aerospace Medical Research Laboratories, Aerospace Medical Division, Air . . . , 1967).
- [18] Bilmes, J. Submodularity in machine learning and artificial intelligence (2022). URL <https://arxiv.org/abs/2202.00132>. 2202.00132.

- [19] Balthazar, C. H. & Scott, C. M. Sentences are key: Helping school-age children and adolescents build sentence skills needed for real language. *American Journal of Speech-Language Pathology* **33**, 564–579 (2024).
- [20] Zipf, G. K. *The psycho-biology of language: An introduction to dynamic philology* (Routledge, 2013).
- [21] Robertson, S. E. *et al.* Okapi at trec-3. *Nist Special Publication Sp* **109**, 109 (1995).